Lecture 6: Statistical Inference

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STAT-S520 Introduction to Statistical Inference

Outline

Point Estimation
- Point Estimator
- Plug-in principle
- Unbiased estimator
- Examples

Hypothesis Testing
- Basic concepts
- Hypothesis Testing for a Population Mean

Set Estimation
- Basic concepts
- Set estimation of the population mean

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Point Estimation

Point Estimator

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $\mu = EX_1$ and $\sigma^2 = \mathrm{var}(X_1) < \infty$

Point Estimation

Point Estimator

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $\mu = EX_1$ and $\sigma^2 = \mathrm{var}(X_1) < \infty$
- An **estimator** is a function of the random sample $(X_1, \ldots, X_n)$, in the way of serving as a reasonable guess for the unknown population attribute (mean, variance, median,.....)
- **An estimator is a random variable because we consider $X_1, \ldots, X_n$ to be randomly generated.**

Point Estimation

Plug-in principle

- In our previous class, we know $\bar{X}_n$ is a good estimator of $\mu = E(X)$ when the sample size n is large enough (LLN and CLT)
- For example, LLN says $\bar{X} \xrightarrow{P} \mu$; This is referred to as the **consistency** of an estimator, meaning that when sample size goes to infinity, eventually you can recover the true value of the parameter
- How to build estimators for other attributes? **Plug-in principle**.

Point Estimation

Plug-in principle

- Let generated data be $(X_1, X_2, \ldots, X_n) = (x_1, x_2, \ldots, x_n)$. Create a discrete probability distribution by assigning a probability of $1/n$ to each of the n observations. This distribution is called the **empirical distribution**.
- Let Y be the random variable such that

$$P(Y = x_i) = \frac{1}{n}, \quad i = 1, 2, \ldots, n.$$

- Then the distribution of Y is the empirical distribution, and write its CDF as $F_n(x)$.

Point Estimation

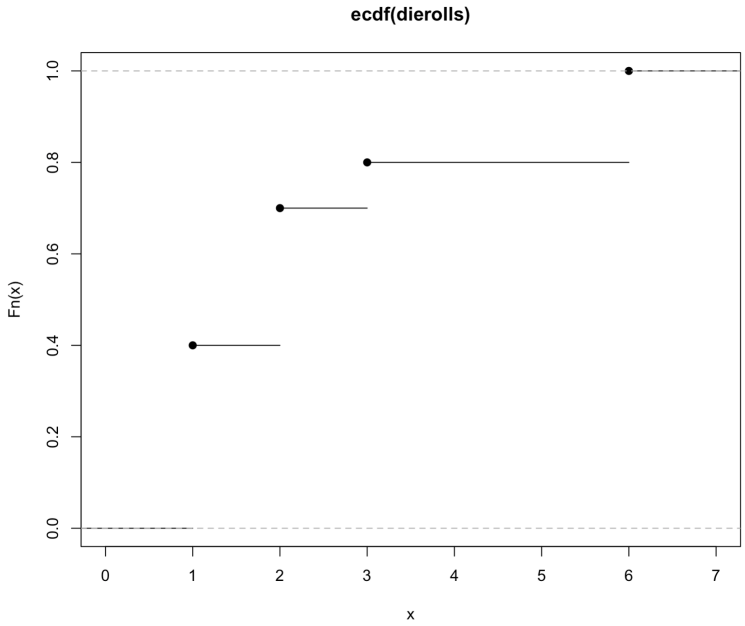
Plug-in principle

I literally rolled a six-sided die ten times, and got:

$$x_1, \ldots, x_{10} = 6,\ 2,\ 2,\ 3,\ 6,\ 2,\ 1,\ 1,\ 1,\ 1$$

The **empirical CDF** is:

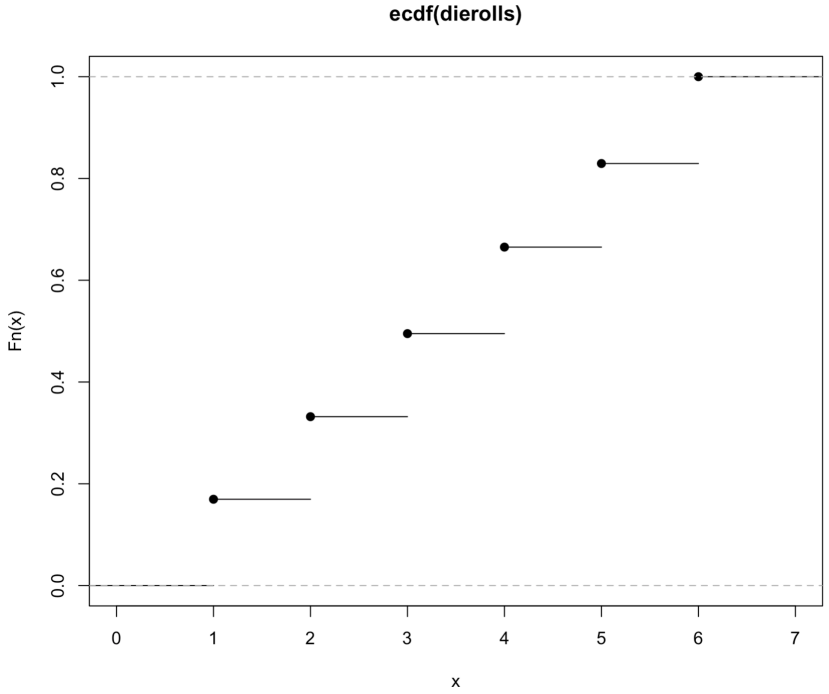
$$F(y) = \begin{cases} 0, & y < 1, \\ 0.4, & 1 \leq y < 2, \\ 0.7, & 2 \leq y < 3, \\ 0.8, & 3 \leq y < 6, \\ 1, & y \geq 6. \end{cases}$$



Point Estimation

Plug-in principle

If we roll the die 10000 times, what will the distribution of Y (empirical CDF) look like? We run the R code (See the canvas for the codes).



- The CDF of Y looks just like the true CDF of each sample $(X_1, \ldots, X_n)$.
- **When n becomes large, F_n gets more and more closed to F. So the distribution of Y gets more and more closed to X as n becomes large.**

Point Estimation

Plug-in principle

- Recall that population attribute is just a function of the true distribution of X ($g(X)$). We could use the corresponding function of Y ($g(Y)$) to estimate $g(X)$.
- Estimator built base on this logic is called the **plug-in estimator**.
- Plug-in mean estimator

$$\mu = E(X) \quad \Leftarrow \quad E(Y) = \sum_{i=1}^{n} x_i P(Y = x_i) = \frac{\sum_{i=1}^{n} x_i}{n}.$$

Take the random generation of samples into account. So

$$\bar{X}_n = \frac{\sum_{i=1}^{n} X_i}{n}$$

is the plug-in estimator of mean.

Point Estimation

Plug-in principle

- Plug-in variance estimator:

$$\sigma^2 = E(X - E(X))^2 \quad \Leftarrow \quad E(Y - EY)^2.$$

We could further write

$$E(Y - EY)^2 = \sum_{i=1}^{n} (x_i - E(Y))^2 P(Y = x_i) = \frac{\sum_{i=1}^{n} \left(x_i - \frac{\sum_{i=1}^{n} x_i}{n} \right)^2}{n}$$

Take the random generation of samples into account. So

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X}_n)^2}{n}$$

is the plug-in estimator of variances.

Point Estimation

Plug-in principle

- The true variances of rolling a fair die:

$$\sigma^2 = \frac{1}{6}(1-3.5)^2 + \frac{1}{6}(2-3.5)^2 + \cdots + \frac{1}{6}(6-3.5)^2 = 2.917$$

- $\hat{\sigma}^2 = 3.833$ with $n = 10$
- $\hat{\sigma}^2 = 2.945$ with $n = 10000$

Point Estimation

Plug-in principle

Given a sample $X_1, X_2, \ldots, X_n$, we can estimate the population quartiles (q_1, q_2, q_3) as follows:

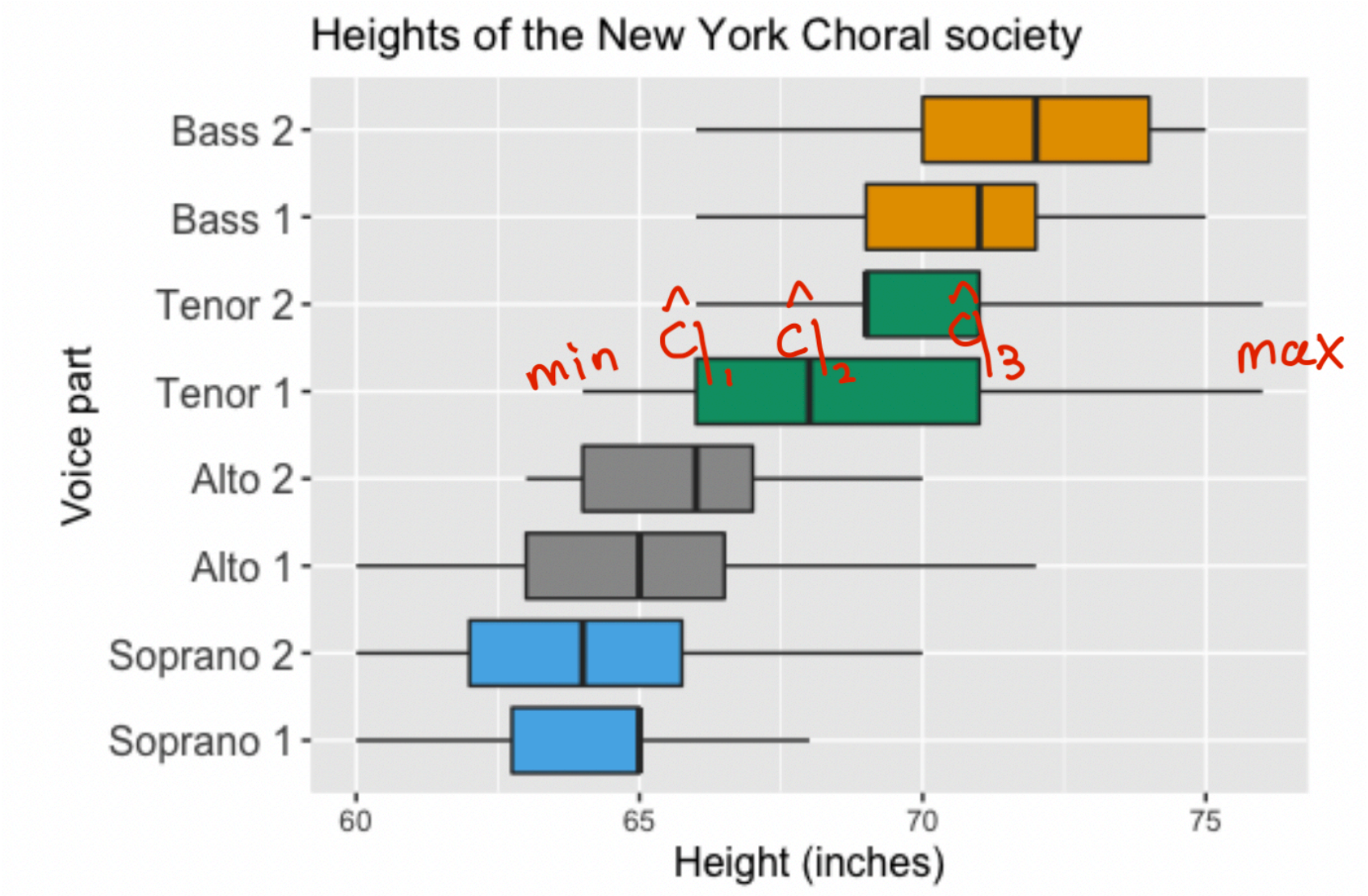
- Sort the data: $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$.
- Estimate the quartiles by their empirical quantiles:

$$\underbrace{X_{(1)}}_{\text{min}} \leq \cdots \leq \underbrace{X_{(0.25n)}}_{\hat{q}_1} \leq \cdots \leq \underbrace{X_{(0.50n)}}_{\hat{q}_2} \leq \cdots \leq \underbrace{X_{(0.75n)}}_{\hat{q}_3} \leq \cdots \leq \underbrace{X_{(n)}}_{\text{max}}.$$

- Plut-in estimator for IQR? $\hat{q}_3 - \hat{q}_1 \Rightarrow q_3 - q_1$.
- We can use the above five quantities to summarize any datasets through a plot—**Box-plot**.

Point Estimation

Plug-in principle



Point Estimation

Unbiased estimator

- An estimator is **unbiased** for the corresponding estimating population attribute, if the expected value always equals the (unknown) population attribute regardless of what the population distribution P is
- Mathematically, let $\widehat{\theta} = T(X_1, \ldots, X_n)$ be an estimator for the (unknown) population attribute θ, where T is a function of the random sample $(X_1, \ldots, X_n)$
- Then $\widehat{\theta}$ is said to be an unbiased estimator for θ, if

$$E\widehat{\theta} = \theta$$

for all possible population distribution P

Point Estimation

Unbiased Estimator

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $\mu = EX_1$ and $\sigma^2 = \mathrm{var}(X_1) < \infty$
- Recall the **Sample mean** is an estimator for the population mean μ

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

- $\bar{X}$ is always unbiased for μ: $E\bar{X} = \mu$ regardless of what P is.

Point Estimation

Unbiased Estimator

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- Recall the **plug-in variance** is given by

$$\widehat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

is an estimator for the population variance σ^2

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- Another estimator for σ^2 is the **Sample variance**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

  is always unbiased for σ^2: $ES^2 = \sigma^2$ regardless of what P is!

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Unbiased Estimator

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- It follows that

$$\widehat{\sigma}^2 = \frac{n-1}{n}S^2$$

  which immediately implies that $\widehat{\sigma}^2$ is not unbiased for σ^2
- We prefer using the sample variance S^2 over the plug-in variance $\widehat{\sigma}^2$

Example

Suppose a certain type of fertilizer has an expected yield per acre of μ_1 with variance σ^2, whereas the expected yield for a second type of fertilizer is μ_2 with the same variance. Let S_1^2 and S_2^2 denote the sample variances of yields based on sample sizes n_1 and n_2, respectively, of the two fertilizers. Is the pooled variance

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

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- Sample variance is unbiased $\implies$ $ES_1^2 = ES_2^2 = \sigma^2$
- Then

$$ES^2 = \frac{(n_1 - 1)ES_1^2 + (n_2 - 1)ES_2^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n)1 + n_2 - 2} = \sigma^2$$

Example

Using a long rod that has length μ, you are going to lay out a square plot in which the length of each side is μ. Thus the area of the plot will be μ^2. However, you do not know the value of μ, so you decide to make n independent measurements $X_1, X_2, \ldots, X_n$ of the length. Assume that each X_i has mean μ (unbiased measurements) and variance σ^2.

(a) Is $\bar{X}^2$ an unbiased estimator for μ^2?

(b) For what value of k is the estimator $\bar{X}^2 - kS^2$ unbiased, where S^2 is the sample variance (which is unbiased for σ^2)?

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Answer:

(a) $\mathrm{var}(\bar{X}) = E\bar{X}^2 - (E\bar{X})^2 = E\bar{X}^2 - \mu^2 > 0 \Longrightarrow E\bar{X}^2 \neq \mu^2$

(b) $E\bar{X}^2 = \mathrm{var}(\bar{X}) + (E\bar{X})^2 = \frac{1}{n}\sigma^2 + \mu^2$

$$E(\bar{X}^2 - kS^2) = E\bar{X}^2 - kES^2 = \frac{1}{n}\sigma^2 + \mu^2 - k\sigma^2 = \mu^2$$

which implies that $k = 1/n$.

Example

Of n_1 randomly selected male smokers, X_1 smoked filter cigarettes, whereas of n_2 randomly selected female smokers, X_2 smoked filter cigarettes. Let p_1 and p_2 denote the probabilities that a randomly selected male and female, respectively, smoke filter cigarettes. Find an unbiased estimator for $p_1 - p_2$.

Example

Suppose the true average growth μ of one type of plant during a 1-year period is identical to that of a second type, but the variance of growth for the first type is σ^2, whereas for the second type, the variance is $4\sigma^2$. Let $X_1, \ldots, X_m$ be m inpendent growth observations on the first type, and let $Y_1, \ldots, Y_n$ be n independent growth observations on the second type.

(a) Let $\delta \in [0, 1]$. Is $\widehat{\mu} = \delta \bar{X} + (1 - \delta)\bar{Y}$ unbiased for μ?

(b) For fixed m and n, find $\mathrm{var}(\widehat{\mu})$.

Example

An investigator wishes to estimate the proportion of students at a certain university who have violated the honor code. Having obtained a random sample of n students, she realizes that asking each, "Have you violated the honor code?" will probably result in some untruthful responses. Consider the following scheme, called a **randomized response** technique. The investigator makes up a deck of 100 cards, of which 50 are of type I and 50 are of type II.

- Type I: Have you violated the honor code (yes or no)?
- Is the last digit of your telephone number a 0, 1, or 2 (yes or no)?

Example

Each student in the random sample is asked to mix the deck, draw a card, and answer the resulting question truthfully. Because of the irrelevant question on type II cards, a yes response no longer stigmatizes the respondent, so we assume that responses are truthful.

Let p denote the proportion of honor-code violators, and $\lambda = P(\text{yes response})$. Then $\lambda = p/2 + 0.15$. Let Y denote the number of yes responses, so $Y \sim \mathrm{Binomial}(n, \lambda)$. Thus Y/n is an unbiased estimator of λ.

Find an unbiased estimator for p based on Y.

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Basic concepts

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Explanation two: This seems to be close to impossible as the probability is $(1/2)^{100}$ is pretty much 0, and in computer simulations it is just 0 (numeric underflow). Do you still believe that your coin is unbiased?

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  - Tossing a coin: $X_1, \ldots, X_n \sim \text{Bernoulii}(p)$ and p determines the distribution of data.
  - Null hypothesis: the coin is fair $\Rightarrow H_0 = \{p \mid p = 1/2\}$
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  - Type II error: wrongly choosing H_0 when the underlying fact is H_1.
- **Significance level:** Typically denoted by α, which is the maximum allowed probability of making the type I error in your test.

- **Significance level and test function:** In the test function formulation, it is required that the probability of the type I error is upper bounded by α:

$$P\left(\phi(X_1, \ldots, X_n) = 1 \mid H_0 \text{ is actually correct}\right) \leq \alpha.$$

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- p**-value:** Under H_0, the significance probability, also referred to as the p-value, is defined to be the probability that the random sample would be **more extreme against H_0 than the observation**.
- **Decision rule:** One first set the significance level $\alpha \in (0, 1)$, which is the upper bound on the maximum type I error probability. The significance level represents the researchers tolerance for wrongly rejecting H_0. Then one make the following decision:

$$\begin{aligned} &\text{Reject } H_0, && \text{if } p < \alpha, \\ &\text{Fail to reject } H_0, && \text{if } p > \alpha. \end{aligned}$$

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- **Significance level:** Different circumstances warrant different choices of α. Conventional choices include 0.05, 0.01, and 0.10.

Example

Story: Andre Drummond used to be a poor free-throw shooter. In a period of time he attempted 100 free throws but only made 32 out of it (These are fake data but sufficiently authentic to reflect his in-game FT performance...). We are interested in knowing whether or not Drummond's free throw percentage equals 0.5.

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Probability model: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and we observe $\sum_{i=1}^{n} x_i = 32$ with $n = 100$. Consider testing $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$ at significance level $\alpha = 0.01$.

Hypothesis Testing

Example

- Consider the test function

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1, & \text{if } \left| \sum_{i=1}^{n} X_i - 50 \right| > c, \\ 0, & \text{otherwise,} \end{cases}$$

- "More extreme than the observed value" means $|\sum_i X_i - 50| > |\sum_i x_i - 50|$

Hypothesis Testing

Example

- Consider the test function

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- "More extreme than the observed value" means $|\sum_i X_i - 50| > |\sum_i x_i - 50|$
- Then p-value is given by

$$\begin{aligned} & P_{H_0}\left(\left|\sum_{i=1}^{n} X_i - 50\right| > |32 - 50|\right) \\ &= P_{H_0}\left(\sum_{i=1}^{n} X_i > 68\right) + P_{H_0}\left(\sum_{i=1}^{n} X_i < 32\right) \\ &= 1 - \texttt{pbinom(68, 100, 0.5)} \\ &\quad + \texttt{pbinom(31, 100, 0.5)} \\ &= 0.0001831432. \end{aligned}$$

- Small p means against H_0, what about large p?
- **Large p-value (I):** A p-value thats not small (e.g. 10% or 20% or 90%) means the data is consistent with the null hypothesis. Any difference between what you observed and what you expect can easily be explained as random error.

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- **Large p-value (II):** However, a large p-value in itself doesnt prove that H_0 is right. In particular, it might that your sample size is too small to prove anything.

Logic of p-value:

- p-value small, large sample: Yichi said: The data would be unusual if the null hypothesis were true. I would say: Be skeptical about your null hypothesis. If your assumptions are met, then this is evidence *for* the alternative hypothesis.
- p-value small, small sample: Yichi said: The data would be unusual if the null hypothesis were true. I would say: Decision might be correct, but highly risky! (but replicate.)
- p-value not small, large sample: The data is consistent with the null hypothesis model.
- p-value not small, small sample: Get more data (basically tells you nothing but fail to reject the null).

Outline

Point Estimation
- Point Estimator
- Plug-in principle
- Unbiased estimator
- Examples

Hypothesis Testing
- Basic concepts
- Hypothesis Testing for a Population Mean

Set Estimation
- Basic concepts
- Set estimation of the population mean

- **Hypothesis testing setup:** Suppose that $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with population mean $\mu = EX_1$ and population variance $\sigma^2 = \mathrm{var}(X_1)$. Consider the problem of testing $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ for some known μ_0.

Hypothesis Testing

Testing for a Population Mean

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- **Test statistic:** A test statistic is used to quantify the null hypothesis.
- When the variance σ^2 is known, we take $|\bar{X}_n - \mu_0|$;
- When the variance σ^2 is unknown, we take $|\bar{X}_n - \mu_0|/S_n$ where S_n^2 is the sample variance

Computing p-value when σ^2 is known:

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$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma},$$

- Compute the observed value of the test statistic

$$z = \frac{\sqrt{n}(\bar{x}_n - \mu_0)}{\sigma},$$

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$$
\begin{aligned}
p\text{-value} &= P_{\mu_0}(|\bar{X}_n - \mu_0| > |\bar{x}_n - \mu_0|) \\
&= 1 - P_{\mu_0}(-|\bar{x}_n - \mu_0| \leq \bar{X}_n - \mu_0 \leq |\bar{x}_n - \mu_0|) \\
&= 1 - P_{\mu_0}\left(-\frac{\sqrt{n}|\bar{x}_n - \mu_0|}{\sigma} \leq \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \leq \frac{\sqrt{n}|\bar{x}_n - \mu_0|}{\sigma}\right) \\
&= 1 - P_{\mu_0}\left(-|z| \leq Z \leq |z|\right) \\
&\approx 1 - [\Phi(|z|) - \Phi(-|z|)] \\
&= \texttt{2pnorm(- abs(z)).}
\end{aligned}
$$

Testing Bernoulli population

- A lot of problems (HWs, Exams, etc.) are Bernoulli
- Random sample $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$.
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- Let $X_1, \ldots, X_n \sim P$; Observe $x_1, \ldots, x_n$; Variance $\sigma^2 = \mathrm{var}(X_1)$ unknown
- In the **test statistic**, replace σ^2 with the sample variance S^2

$$T = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S},$$

- Compute the observed value of the test statistic

$$t = \frac{\sqrt{n}(\bar{x}_n - \mu_0)}{s},$$

- Compute the p-value using CLT:

$$p\text{-value} = 2 \texttt{ * pnorm(- abs(t)).}$$

Sample variance of Bernoulli population

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- **Answer:** Use the fact that $x_i^2 = x_i$ (because it is either 0 or 1)!
- **Formula for sample variance:** Compute sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + n\bar{x}^2\right)$$

$$= \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i - 2n\bar{x}^2 + n\bar{x}^2\right) = \frac{1}{n-1}(n\bar{x} - n\bar{x}^2) = \frac{n}{n-1}\bar{x}(1-\bar{x})$$

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Testing for a Population Mean: Example

- Assume the population variance is unknown
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$$s^2 = \frac{n}{n-1}\bar{x}(1-\bar{x}) = \frac{100}{99} \times \frac{32}{100}\left(1 - \frac{32}{100}\right) = 0.2198.$$

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- **Draw conclusion:** We reject the null hypothesis that $p = 0.5$.

- Assume the population variance is known:

$$\sigma^2 = p_0(1 - p_0) = 0.5 \times 0.5 = 0.25.$$

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Hypothesis Testing

Testing for a Population Mean: One-sided problems (variance known)

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(a) $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$; Already discussed.

(b) $H_0 : \mu \leq \mu_0$ against $H_1 : \mu > \mu_0$; This typically happens when you observe a sample mean $\bar{x}$ that is significantly larger than μ_0 such that you are suspicious about the null H_0. The p-value is computed according to the direction of H_1 :

$$p\text{-value} = P_{\mu_0}\left\{\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} > \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma}\right\} = \texttt{1 - pnorm(z)}$$

Typically $z > 0$. If $z \leq 0$ then the p-value must be at least 0.5.

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One-sided problem: Example

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2. **Compute the sample variance:**

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3. **Compute the statistic:**

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4. **Compute the p-value:**

$$p\text{-value} = \texttt{pnorm(-3.8394)} = 6.1668 \times 10^{-5} < \alpha.$$

5. **Draw conclusion:** We reject the null hypothesis that $p \geq 0.5$.

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$$p\text{-value} = \texttt{pnorm(-3.6)} = 0.0001591 < \alpha.$$

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Example

Story: The shooting guard Duncan Robinson from Miami Heat is one of the elite pure shooters in NBA (at least most people agree with that?). In the 2019-2020 NBA regular season, Duncan Robinson has participated in 73 games, averaging 8.3 3-point attempts (3PA) per-game and made 3.7 out of 8.3.

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`https://www.basketball-reference.com/players/r/robindu01.html`

A critic does not agree with the comment that Duncan Robinson is an elite shooter and is interested in knowing whether or Duncan Robinson's 3-piont field goal percentage is no greater than 40%.

Hypothesis Testing

Example: Formulating the correct probability model for testing (variance unknown)

1. **Probability model:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and we observe $\sum_{i=1}^{n} x_i \approx 3.7 \times 73 \approx 270$ with $n \approx 8.3 \times 73 \approx 606$. Consider testing $H_0 : p \leq 0.4$ against $H_1 : p > 0.4$.

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2. **Choose a significance level:** Here we choose $\alpha = 0.05$ but the α's in the problems you will see in HW and Exams given!

Hypothesis Testing

Example: Formulating the correct probability model for testing (variance unknown)

1. **Probability model:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and we observe $\sum_{i=1}^{n} x_i \approx 3.7 \times 73 \approx 270$ with $n \approx 8.3 \times 73 \approx 606$. Consider testing $H_0 : p \leq 0.4$ against $H_1 : p > 0.4$.
2. **Choose a significance level:** Here we choose $\alpha = 0.05$ but the α's in the problems you will see in HW and Exams given!
3. Assume the population variance is unknown
4. **Computation of the statistic:**

$$s^2 = \frac{1}{n-1}\left\{\sum_{i=1}^{n} x_i - n(\bar{x})^2\right\} = \frac{1}{606} \times \left\{270 - 606 \times \left(\frac{270}{606}\right)^2\right\} = 0.2470$$

$$t = \frac{\sqrt{n}(\bar{x} - p_0)}{s} = \frac{\sqrt{606} * (270/606 - 0.4)}{\sqrt{0.2470}} = 2.2559$$

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5. **Computation of the p-value for the one-sided test:**

$$p\text{-value} = P_{p_0}(T > t) = 1 \texttt{ - pnorm(2.2559)} = 0.01203845 < \alpha.$$

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6. **Draw conclusion:** We reject the null hypothesis $H_0 : p \leq 0.4$.

Hypothesis Testing

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$$\sigma^2 = p_0(1 - p_0) = 0.4 \times (1 - 0.4) = 0.24.$$
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$$z = \frac{\sqrt{n}(\bar{x} - p_0)}{\sigma} = \frac{\sqrt{606} * (270/606 - 0.4)}{\sqrt{0.24}} = 2.2886$$

Hypothesis Testing

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Hypothesis Testing

Textbook Problems that are related

- **Section 9.6:** 1, 2, 3, 4, 5, 6, 7, 12(a)
- **Homework problems:** 6, 7, 12(a) (to be formally assigned on Canvas)

Hypothesis Testing

Example

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Poi}(\lambda)$. When n is large, the central limit theorem tells you that

$$Z = \frac{\sqrt{n}(\bar{X} - \lambda)}{\sqrt{\lambda}}$$

has approximately a standard normal distribution. For testing $H_0 : \lambda = \lambda_0$, we can replace λ by λ_0 in the equation for Z instead of S^2. Note that this test is specifically tailored for the Poisson population.

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If the number of requests for consulting received by a certain statistician during a 5-day work week has a Poisson distribution and the total number of consulting requests during a 36-week period is 160, does this suggest that the true average number of weekly requests equals 4.0? Test at significance level $\alpha = .02$.

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Point Estimation
- Point Estimator
- Plug-in principle
- Unbiased estimator
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Set Estimation

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  - If H_0 is not rejected, then ψ_0 is plausible.

Set Estimation

Basic concepts

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- This is referred to as the **duality** between testing and confidence interval.
- This formulation of confidence interval is not very useful in practice
- In practice we use the following formulation of the confidence interval
- **A $1-\alpha$ confidence interval (CI)** for the population attribute $\theta = \psi(P)$ (sometimes referred to as a parameter), is an interval of the form (A, B), where A, B are random variables, s.t.

$$P(A \leq \theta \leq B) = 1 - \alpha.$$

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Set estimation of the population mean

Confidence interval of mean with σ^2 known

- **Note:** Hardly useful in practice but is the gateway to σ^2 being uknown
- The more useful and friendly version of the formulation of CI is given below
- Now we assume that $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$
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$$I = \left(\bar{X}_n - q\frac{\sigma}{\sqrt{n}}, \bar{X}_n + q\frac{\sigma}{\sqrt{n}} \right)$$

where $q =$ `qnorm(1 - alpha/2)` is referred to as the **critical value**

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- **Note:** In HW and Exam problems, you are typically given observed random samples; Replace the estimator $\bar{X}$ with the observed value $\bar{x}$ instead and compute the specific numbers!

Set estimation of the population mean

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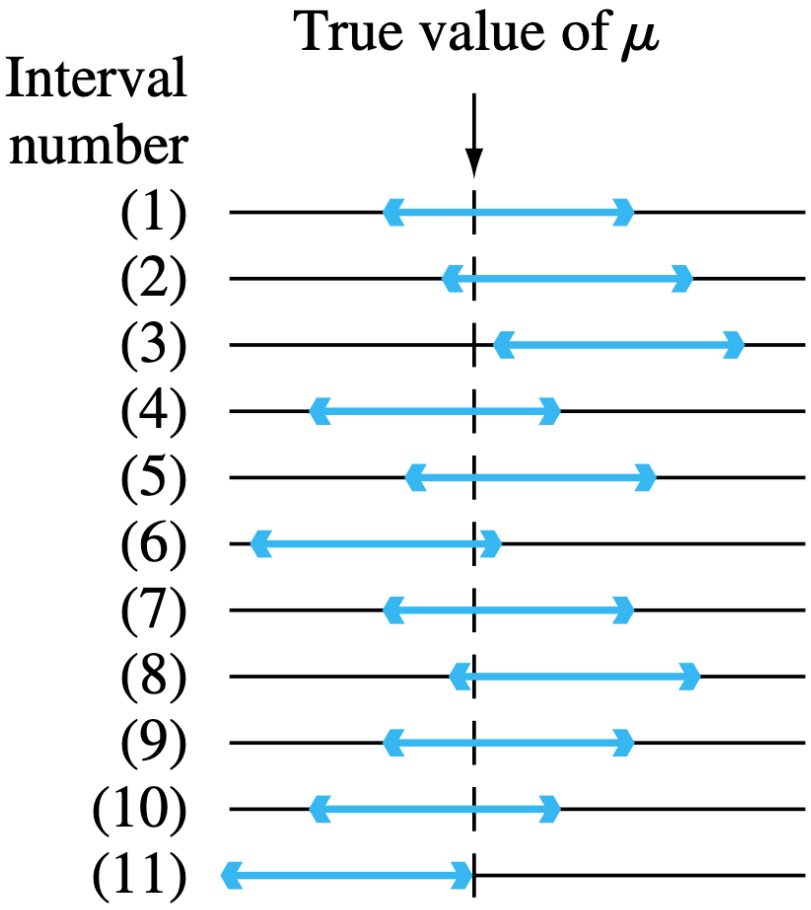
Set estimation of the population mean

Confidence interval: What does it mean?



Set estimation of the population mean

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Set estimation of the population mean

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- These results rely on various CLT for various estimators
- More results in STAT-S420/620, STAT-S721, STAT-S722, etc.

Set estimation of the population mean

Textbook problems that are related

- **Section 9.6:** 8, 9, 10, 11
- **Homework problems:** 8, 9, 10, 11 (to be assigned formally on Canvas)

Example

(Thanks to Prof. Michael Trosset for making it fun!) Let p denote the probability that LeBron James will hit a wide-open 10-foot jump shot. To estimate p, a basketball fan finds video recordings of 105 occasions on which James attempted such a shot and finds that he was successful on 60 of them. (These are just fake data for illustration). Construct a 0.95-level confidence interval for p

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- **Compute the sample variance:**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{104}\left(\sum_{i=1}^{n} x_i - 105\bar{x}^2\right) = 0.2472.$$

Set estimation of the population mean

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- **Compute the confidence interval:**

$$\left(\bar{x} - \texttt{qnorm(0.975)}\,\frac{s}{\sqrt{n}}, \bar{x} + \texttt{qnorm(0.975)}\,\frac{s}{\sqrt{n}}\right) = (0.4763, 0.6665)$$

Example

Story: The shooting guard Duncan Robinson from Miami Heat is one of the elite pure shooters in NBA (at least most people agree with that?). In the 2019-2020 NBA regular season, Duncan Robinson has participated in 73 games, averaging 8.3 3-point attempts (3PA) per-game and made 3.7 out of 8.3.

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`https://www.basketball-reference.com/players/r/robindu01.html`

Find the 95% confidence interval of the 3-point field goal percentage of Duncan Robinson

Set estimation of the population mean

Example of confidence interval

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$$\left(\bar{x} - \texttt{qnorm(0.975)}\frac{s}{\sqrt{n}}, \bar{x} + \texttt{qnorm(0.975)}\frac{s}{\sqrt{n}}\right)$$
$$= (0.4059751, 0.485114)$$

Example (**Important!**)

Consider $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we obtain the confidence interval for $\mu = EX_1$. Assume the variance $\sigma^2 = \mathrm{var}(X_1)$ is known. Suppose the sample size n is large enough so that the central limit theorem is applicable.

- How much do we need to increase the sample size n if we need the width of the confidence interval to be halved?
- If the sample size is increased by a factor of 25, what effect will this have on the width of the interval?

Set Estimation

Example

Consider the next 1000 95% confidence intervals for μ that a statistical consultant will obtain for various clients. Suppose the data sets on which the intervals are based are selected independently of one another

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Consider the next 1000 95% confidence intervals for μ that a statistical consultant will obtain for various clients. Suppose the data sets on which the intervals are based are selected independently of one another

(a) How many of these 1000 intervals do you expect to capture the corresponding value of μ?

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Answer: 950, .8714 if using CLT; 0.8731 if using `pbinom`.

Example

The article *An Evaluation of Football Helmets Under Impact Conditions* (Amer. J. Sports Medicine, 1984: 233237) reports that when each football helmet in a random sample of 37 suspension-type helmets was subjected to a certain impact test, 24 showed damage. Let p denote the proportion of all helmets of this type that would show damage when tested in the prescribed manner.

- Find 99% confidence interval for p.
- What sample size would be required for the width of a 99% confidence interval to be at most .10, irrespective of $\widehat{p}$?

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Answer:

- (0.4437017, 0.8535956)
- 657